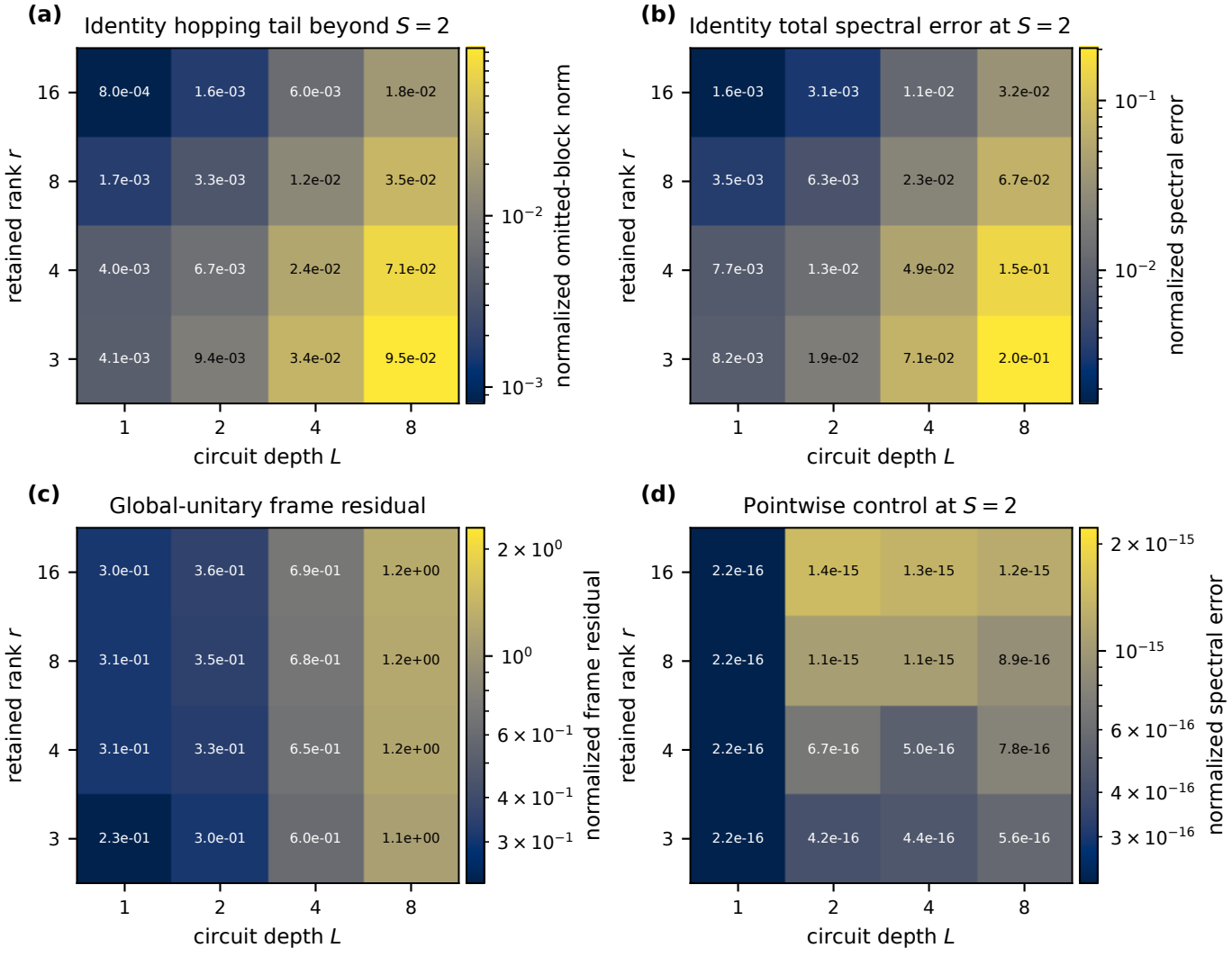
(a)
Identity hopping tail beyond $S = 2$
retained rank r
16
8
4
3
8.0e-04
1.6e-03
6.0e-03
1.8e-02
1.7e-03
3.3e-03
1.2e-02
3.5e-02
4.0e-03
6.7e-03
2.4e-02
7.1e-02
4.1e-03
9.4e-03
3.4e-02
9.5e-02
1
2
4
8
circuit depth L
10^{-2}
10^{-3}
normalized omitted-block norm
(b)
Identity total spectral error at $S = 2$
retained rank r
16
8
4
3
1.6e-03
3.1e-03
1.1e-02
3.2e-02
3.5e-03
6.3e-03
2.3e-02
6.7e-02
7.7e-03
1.3e-02
4.9e-02
1.5e-01
8.2e-03
1.9e-02
7.1e-02
2.0e-01
1
2
4
8
circuit depth L
10^{-1}
10^{-2}
normalized spectral error
(c)
Global-unitary frame residual
retained rank r
16
8
4
3
3.0e-01
3.6e-01
6.9e-01
1.2e+00
3.1e-01
3.5e-01
6.8e-01
1.2e+00
3.1e-01
3.3e-01
6.5e-01
1.2e+00
2.3e-01
3.0e-01
6.0e-01
1.1e+00
1
2
4
8
circuit depth L
2×10^{0}
10^{0}
6×10^{-1}
4×10^{-1}
3×10^{-1}
normalized frame residual
(d)
Pointwise control at $S = 2$
retained rank r
16
8
4
3
2.2e-16
1.4e-15
1.3e-15
1.2e-15
2.2e-16
1.1e-15
1.1e-15
8.9e-16
2.2e-16
6.7e-16
5.0e-16
7.8e-16
2.2e-16
4.2e-16
4.4e-16
5.6e-16
1
2
4
8
circuit depth L
2×10^{-15}
10^{-15}
6×10^{-16}
4×10^{-16}
3×10^{-16}
normalized spectral error